Two Dimensional Array

13 July 2026 16:45

A two dimensional (2D) array is an array whose elements are themselves arrays. It is commonly used to represent data in the form of rows and columns, just like a table, spreadsheet, chess board, sitting arrangement or matrix.

For example:

Roll no	Name	Marks
101	Amit	89
102	Priya	95
103	Rahul	78

This tabular data can be represented using a 2D array.

Declaration of a 2D Array
SyntaxL

```
dataType[][] arrayName;
      Or
dataType arrayName[][];
```

Example:

```
int[][] marks;
```

This statement only declares the array reference, no memory is allocated.

Creating a 2D Array
Memory is allocated using the **new** keyword.

Syntax:

```
Arrayname = new dataType[rows][columns];
```

Example:

```
marks = new dataType[3][4];
      Or
int[][] marks = new dataType[3][4];
```

This creates a 2D array with:

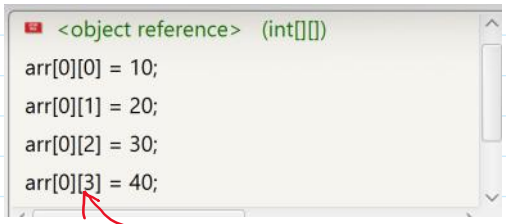
- **3 rows**
- **4 columns**
- **Total elements = 3 * 4 = 12**

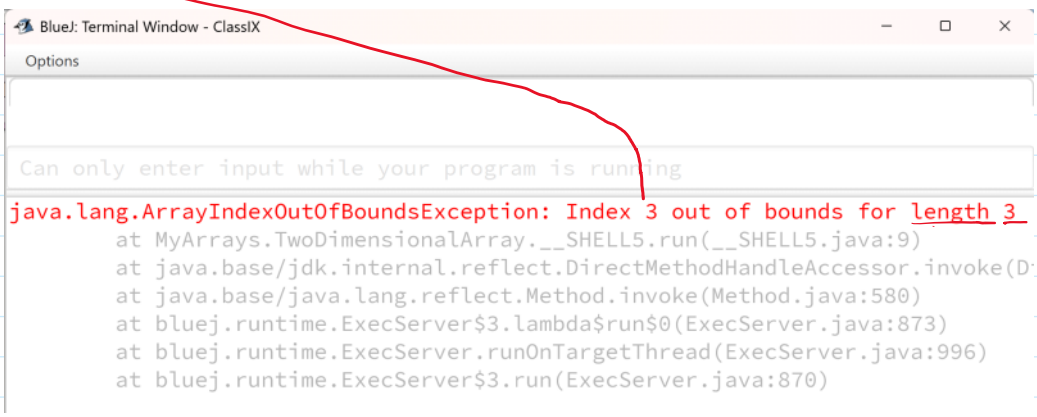
Memory representation:

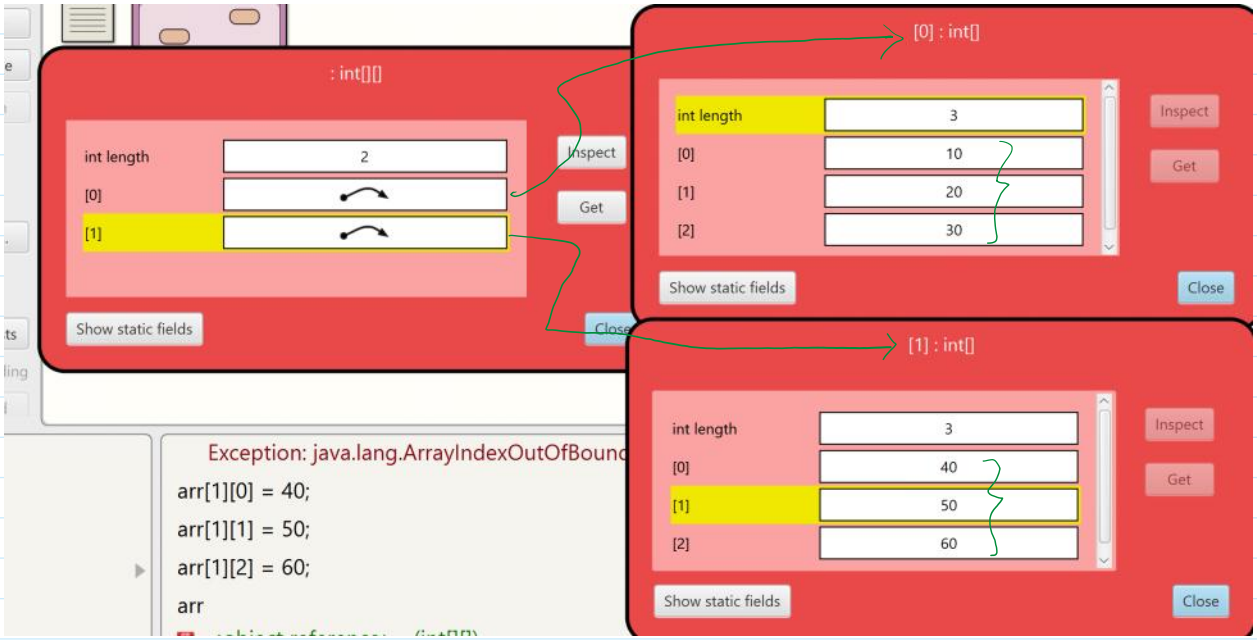
	0,0	0,1	0,2	0,3
	1,0	1,1	1,2	1,3
	2,0	2,1	2,2	2,3

Initializing a 2D array

Method 1: Using Assignment





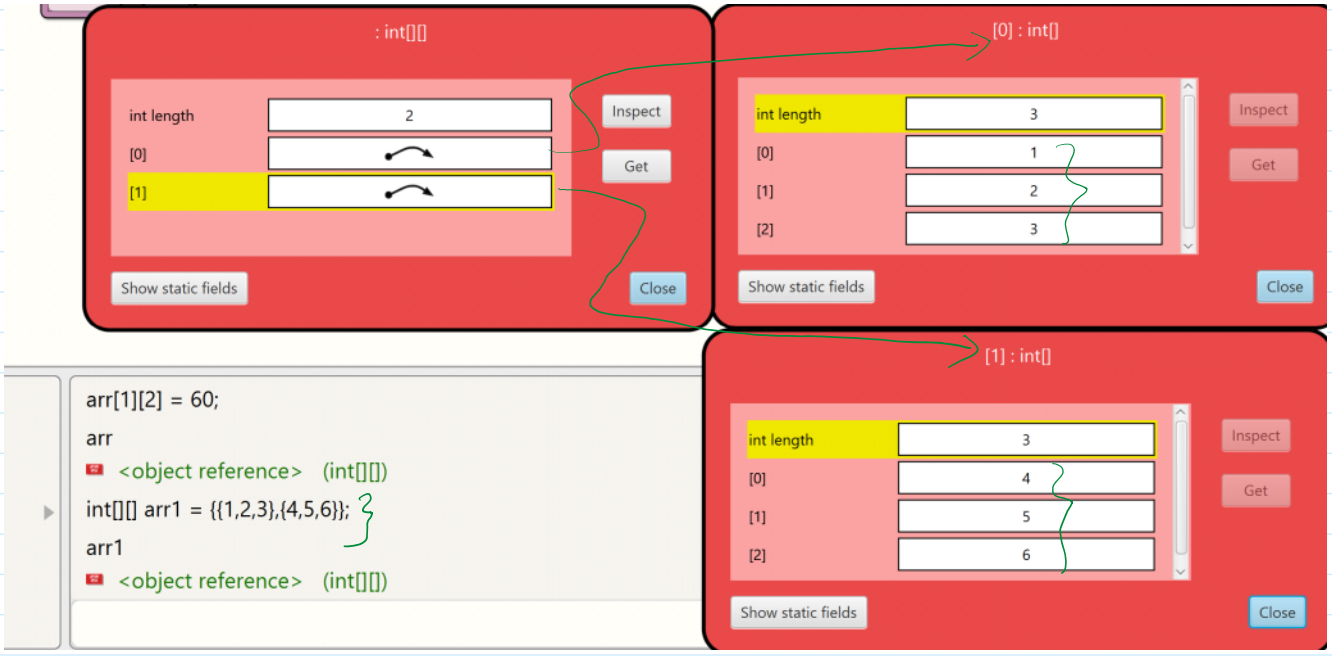


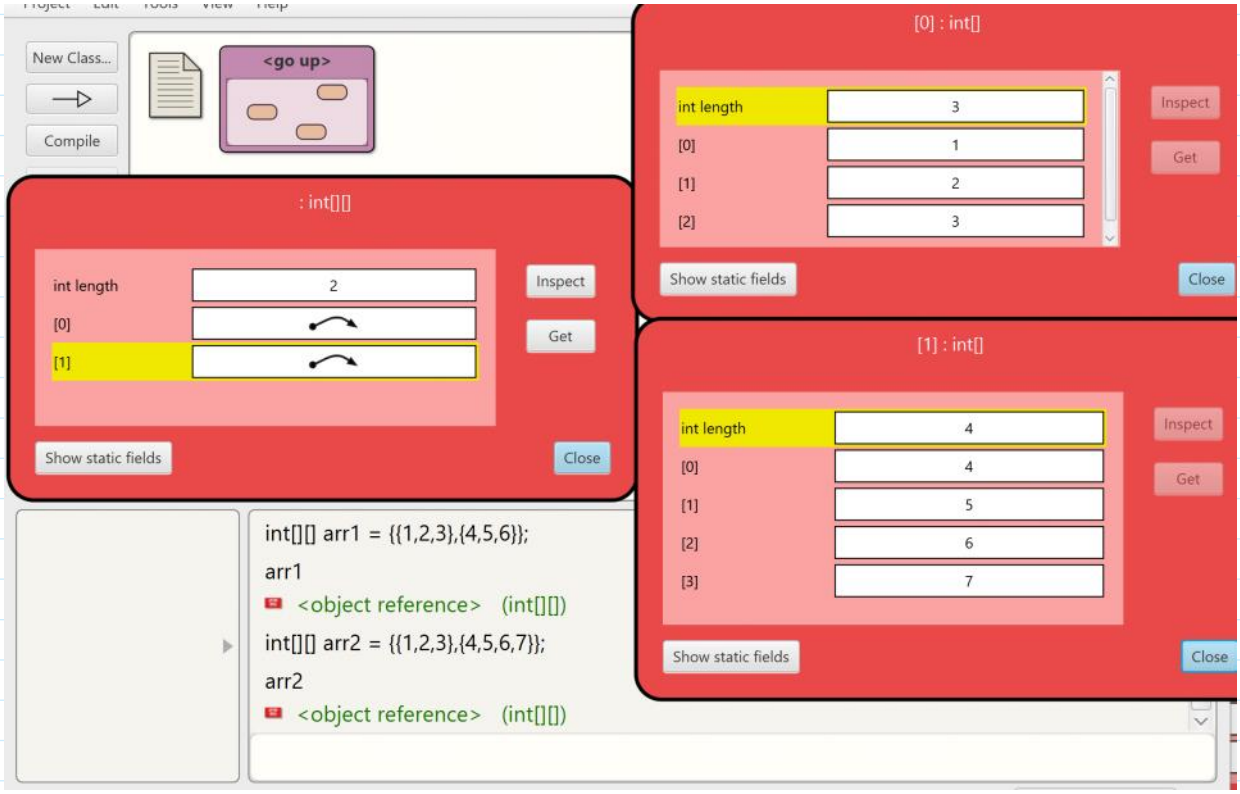

```
int[][] arr = new int[2][3];
arr
<object reference>   (int[][])
arr[0][0] = 10;
arr[0][1] = 20;
arr[0][2] = 30;
arr[0][3] = 40;
Exception: java.lang.ArrayIndexOutOfBoundsException (Index 3 out of bounds for
length 3)
arr[1][0] = 40;
arr[1][1] = 50;
```

```
arr[1][2] = 60;
arr
<object reference>   (int[][])
```

Method 2: Using Array Initializer

```
int[][] arr = {
   {10, 20, 30},
   {40, 50, 60}
};
```





Accessing Elements

Syntax:

```
arrayName[row][column];
```

Example:

```
System.out.print("\n"+arr2[0][2]);
```

Output:

```
3
```

Reading Values from the User

```
package MyArrays.TwoDimensionalArray;
import java.util.*;


/**
 * Write a description of class TwoDArrayUtil here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class TwoDArrayUtil
{
    private static Scanner sc = new Scanner(System.in);
    /**
     * This function takes data input in each element from the user.
     * @param arr int array in which data will inserted by the user
     * @param msg Any message or text displayed on the terminal.
     */
```

```
    public static void inputInt2DArray(int[][] arr, String msg){
        System.out.print("\n" + msg + "\n");
        int i, j;
        for(i = 0; i < arr.length; i++){
            for(j = 0; j < arr[i].length; j++){
                System.out.print("Enter integer value in ["+i+"]["+j+"]: ");
                arr[i][j] = sc.nextInt();
            }
        }
    }

    /**
     * This functions display on terminal int 2D array in matrix format
     * @param arr Two dimensional int array.
     * @param msg Any String message print with Array.
     */
    public static void printInt2DArray(int[][] arr, String msg){
        System.out.print("\n" + msg + "\n");
        int i, j;
        for(i = 0; i < arr.length; i++){
            for(j = 0; j < arr[i].length; j++){
                System.out.print("\t" + arr[i][j]);
            }
            System.out.print("\n");
        }
    }
}

package MyArrays.TwoDimensionalArray;
import java.util.*;


/**
 * Write a description of class Test2DArray here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Test2DArray
{
    public static void main(String[] args){
        int rows, columns;
        Scanner sc = new Scanner(System.in);
        System.out.print("\fEnter number of rows: ");
        rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        columns = sc.nextInt();
        int[][] A = new int[rows][columns];
        TwoDArrayUtil.inputInt2DArray(A, "Enter values in A[][]:");
        TwoDArrayUtil.printInt2DArray(A, "Array A[][]: ");
    }
}
```

number of rows in 2-D array

number of columns in 2-D array

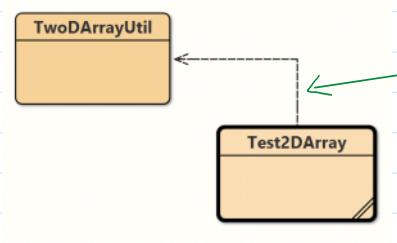
Traversing of Array

Output:

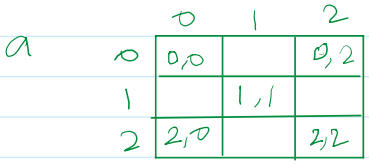
```
BlueJ: Terminal Window - ClassIX
Options

Enter values in A[][]:
Enter integer value in [0][0]: 10
Enter integer value in [0][1]: 12
Enter integer value in [0][2]: 14
Enter integer value in [1][0]: 16
Enter integer value in [1][1]: 18
Enter integer value in [1][2]: 20

Array A[][]:
        10      12      14
        16      18      20
```



Test2DArray class containing the reference of TwoDArrayUtil class. This concept is called Containership in OOP



a[i][i] ← left diagonal

a[i][a.length-1-i] ← right diagonal

Array Questions in Java

1. **Matrix Addition** Write a program to input two matrices of size N x M. Perform element-wise addition and print the resulting matrix in tabular form. Extend the program to also compute the difference of the matrices.

2. **Matrix Multiplication** Accept two matrices A[m x n] and B[n x p]. Multiply them to produce matrix C[m x p]. Validate dimensions before multiplication. Print the result neatly.

3. **Transpose of Matrix** Input a matrix and print its transpose. Extend the program to check if the matrix is symmetric (A[i][j] == A[j][i]).

4. **Diagonal Operations** For a square matrix, compute:
   - Sum of main diagonal
   - Sum of secondary diagonal
   - Difference between them

5. **Spiral Traversal** Input a matrix and print its elements in spiral order

(clockwise). Extend to anti-clockwise spiral traversal.

6. **Snake Pattern Traversal** Print elements row by row, alternating left-to-right and right-to-left. Example:
   Code
   1 2 3
   6 5 4
   7 8 9
7. **Row with Maximum Sum** Find the row with the maximum sum. Print the row index, sum, and elements of that row.

8. **Column with Maximum Sum** Find the column with the maximum sum. Print the column index, sum, and elements of that column.

9. **Search Element** Input a matrix and search for a given element. Print its position (row, column). If not found, display “Element not present”.

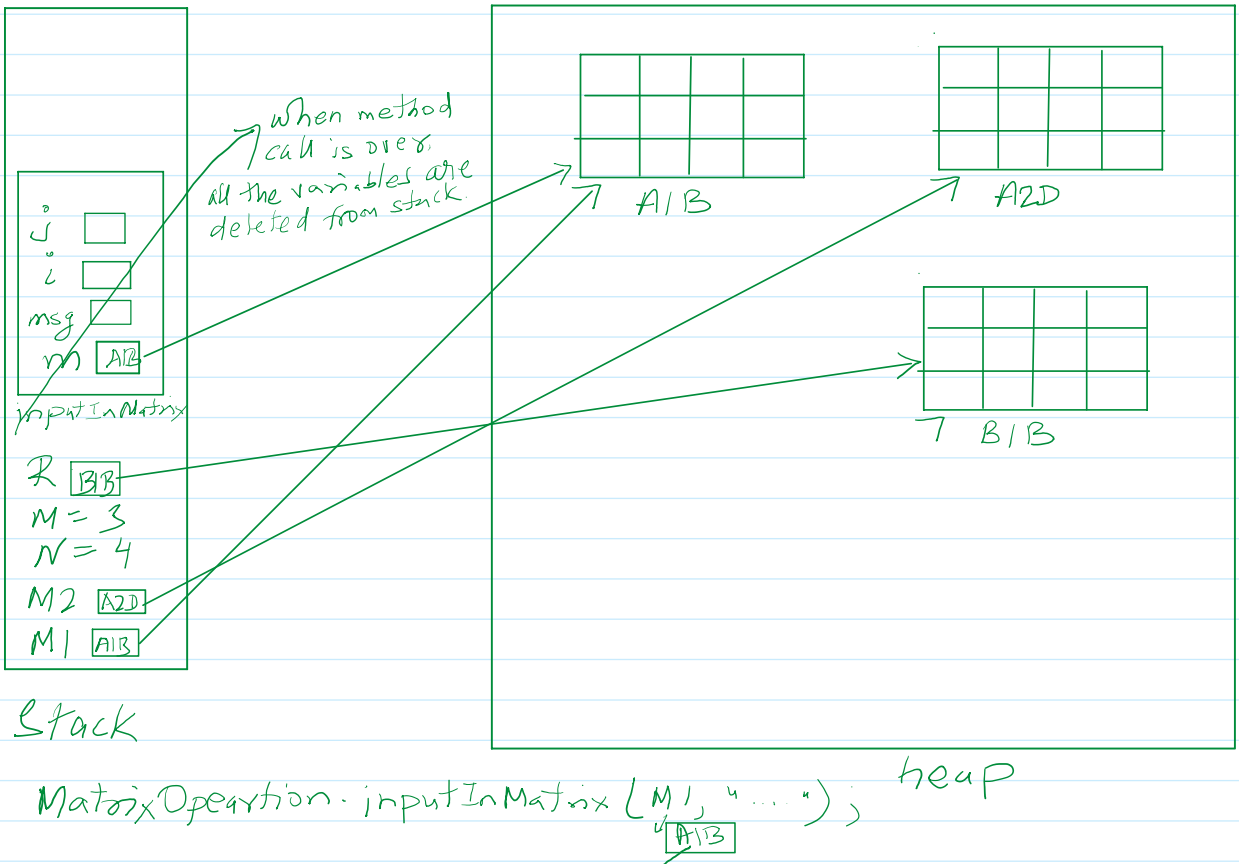
10. **Rotate Matrix 90°** Rotate a square matrix clockwise by 90°. Extend the program to rotate by 180° and 270°.

11. **Boundary Traversal** Print only the boundary elements of a matrix. Example:
    Code
    1 2 3
    4 5
    6 7 8
12. **Magic Square Check** Input a square matrix. Check if it is a magic square (sum of rows, columns, and diagonals are equal).

13. **Pascal’s Triangle** Generate Pascal’s Triangle using a 2D array. Print it in triangular format.

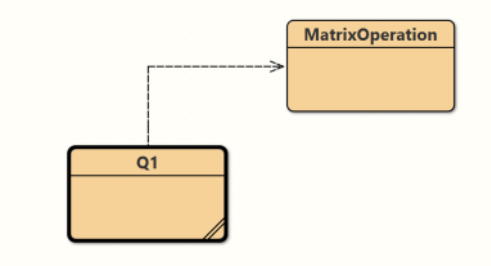
14. **Sudoku Validator** Input a 9x9 Sudoku grid. Validate if:
    - Each row contains digits 1–9 without repetition.
    - Each column contains digits 1–9 without repetition.
    - Each 3x3 subgrid contains digits 1–9 without repetition.

15. **Image Pixel Manipulation** Represent a grayscale image as a 2D array (values 0–255). Implement:
    - Brightness increase/decrease.
    - Horizontal flip.
    - Vertical flip.
    - Inversion (255 – value).



MatrixOperation.inputInMatrix(M1, "....");

```
public static void inputInMatrix(int m[][], String msg){
        System.out.print("\n"+msg+"\n");
        int i, j;
        for(i = 0; i < m.length; i++){
            for(j = 0; j < m[i].length; j++){
                System.out.print("Enter integer value in ["+i+"]["+j+"]: ");
                m[i][j] = sc.nextInt();
            }
        }
    }
```




```
package MyArrays.TwoDimensionalArray;
import java.util.*;

/**
 *
 * @author (your name)
 * @version (a version number or a date)
 */
```

```
public class MatrixOperation
{
    public static Scanner sc = new Scanner(System.in);

    public static int[][] add(int m1[][], int m2[][]){
        int R[][] = new int[m1.length][m1[0].length];
        int i, j;
        for(i = 0; i < m1.length; i++){
            for(j = 0; j < m2[i].length; j++){
                R[i][j] = m1[i][j] + m2[i][j];
            }
        }
        return R;
    }

    public static int[][] subtract(int m1[][], int m2[][]){
        int R[][] = new int[m1.length][m1[0].length];
        int i, j;
        for(i = 0; i < m1.length; i++){
            for(j = 0; j < m2[i].length; j++){
                R[i][j] = m1[i][j] - m2[i][j];
            }
        }
        return R;
    }

    public static void inputInMatrix(int m[][], String msg){
        System.out.print("\n"+msg+"\n");
        int i, j;
        for(i = 0; i < m.length; i++){
            for(j = 0; j < m[i].length; j++){
                System.out.print("Enter integer value in ["+i+"]["+j+"]: ");
                m[i][j] = sc.nextInt();
            }
        }
    }

    public static void printMatrix(int m[][], String msg){
        System.out.print("\n"+msg+"\n");
        int i, j;
        for(i = 0; i < m.length; i++){
            for(j = 0; j < m[i].length; j++){
                System.out.print("\t"+m[i][j]);
            }
            System.out.print("\n");
        }
    }
}

package MyArrays.TwoDimensionalArray;
import java.util.*;
```

```
/**
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q1
{
    public static Scanner sc = new Scanner(System.in);
    public static void main(String args[]){
        System.out.print("\f");
        int M, N, M1[][], M2[][], R[][];
        System.out.print("Enter number of rows: ");
        M = sc.nextInt();
        System.out.print("Enter number of columns: ");
        N = sc.nextInt();
        //memory allocation for Array
        M1 = new int[M][N];
        M2 = new int[M][N];
        R = new int[M][N];

        MatrixOperation.inputInMatrix(M1, "Enter values in M1[][]: ");
        MatrixOperation.inputInMatrix(M2, "Enter values in M2[][]: ");
        R = MatrixOperation.add(M1, M2);
        MatrixOperation.printMatrix(M1, "First matrix:");
        MatrixOperation.printMatrix(M2, "Second matrix:");
        MatrixOperation.printMatrix(R, "Sum of first matrix and second matrix:");
        R = MatrixOperation.subtract(M1, M2);
        MatrixOperation.printMatrix(R, "Difference of first matrix and second matrix:");
    }
}
```

Output:

```
BlueJ: Terminal Window - ClassIX
Options
Enter number of rows: 2
Enter number of columns: 2

Enter values in M1[][]:
Enter integer value in [0][0]: 1
Enter integer value in [0][1]: 2
Enter integer value in [1][0]: 3
Enter integer value in [1][1]: 4

Enter values in M2[][]:
Enter integer value in [0][0]: 5
Enter integer value in [0][1]: 6
Enter integer value in [1][0]: 7
Enter integer value in [1][1]: 8

First matrix:
        1       2
        3       4

Second matrix:
        5       6
        7       8

Sum of first matrix and second matrix:
        6       8
        10      12

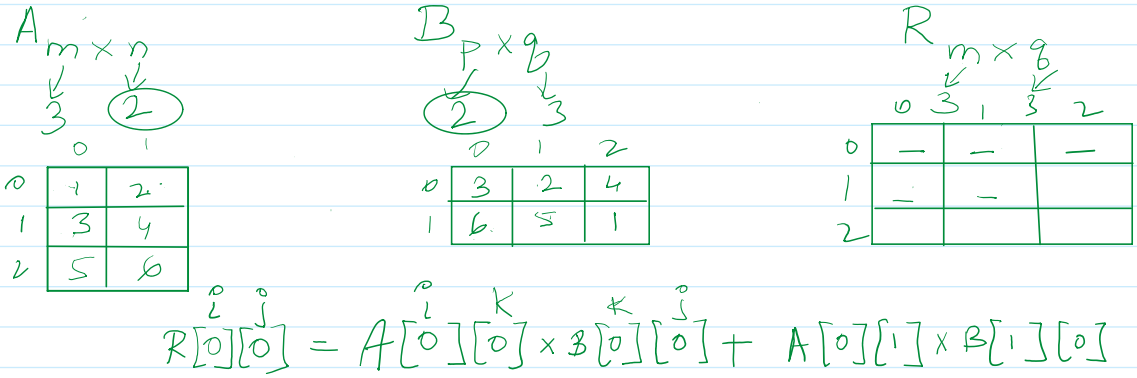
Difference of first matrix and second matrix:
        -4      -4
        -4      -4
```

Matrix multiplication

To multiply two matrices. The number of columns in the first matrix must be equal the number of rows in the second matrix. (M X P and P x N yields M X N). Each element in the result is a dot product of a row from the first matrix and a column from the 2nd matrix.

The rules of matrix multiplication

- **Dimension Match:** Columns of matrix A must equals rows of matrix B.
- **Result size:** if A is m x p and B is p x n, product AB is m x n.
- **Not Commutative: A x B is usually not equal to B x A**
- **Dot Product:** Multiply matching row elements by column elements and then add them.



$R[i][j] = A[i][k] \times B[k][j] + A[i][k] \times B[k][j]$

$R[0][0] = A[0][0] \times B[0][0] + A[0][1] \times B[1][0]$

$R[0][1] = A[0][0] \times B[0][1] + A[0][1] \times B[1][1]$

$R[0][2] = A[0][0] \times B[0][2] + A[0][1] \times B[1][2]$

$R[1][0] = A[1][0] \times B[0][0] + A[1][1] \times B[1][0]$

$R[1][1] = A[1][0] \times B[0][1] + A[1][1] \times B[1][1]$

```
for(i=0; i<m; i++){
    for(j=0; j<q; j++){
        for(k=0; k<n; k++){
            R[i][j] += A[i][k] * B[k][j];
        }
    }
}
```

i = 0 1

j = 0 1 2

k = 0 1 2 0 1 2, 0

$0 + 1 * 3$, $0 + 1 * 2$

$3 + 2 * 6$ $2 + 2 * 5$

A $m \times n$ (3 × 2)

	0	1
0	1	2
1	3	4
2	5	6

B $p \times q$ (2 × 3)

	0	1	2
0	3	2	4
1	6	5	1

R $m \times q$ (3 × 3)

	0	1	2
0	15	12	
1			
2			

A

1	4	5
4	2	6
5	6	3

	0	1	2	3
0	1	5	6	7
1	5	2	8	9
2	6	8	3	10
3	7	9	10	4

$[0][2] = = [2][0]$

$[i][j] = = [j][i]$

A =

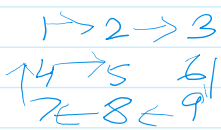
1	2	3
4	5	6
7	8	9

$\Rightarrow A_T =$

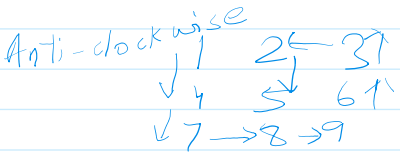
1	4	7
2	5	8
3	6	9

$A = A_T$ [Symmetric matrix]

1. **Spiral Traversal** Input a matrix and print its elements in spiral order (clockwise). Extend to anti-clockwise spiral traversal.



top row (left to right) : 1 2 3
right column (Top to bottom) : 6 9
bottom row (Right to left) : 8 7
left column (Bottom to top) : 4
Inner row (Left to right) : 5



Final output : 1 2 3 6 9 8 7 4 5

```
package MyArrays.TwoDimensionalArray;
import java.util.*;

/**
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class SpiralTraversal
{
    public static Scanner sc = new Scanner(System.in);
    public static void main(String args[]){
        System.out.print("\fEnter the dimension of 2d matrix: ");
        int m = sc.nextInt(), n = sc.nextInt();
        int matrix[][] = new int[m][n];
        Random r = new Random();
        int i, j;
        for(i = 0; i < m; i++){
            for(j = 0; j < n; j++){
                matrix[i][j] = r.nextInt(10, 100);
            }
        }

        TwoDArrayUtil.printInt2DArray(matrix, "Matrix:");
        printClockwise(matrix, m, n);
    }

    public static void printClockwise(int[][] matrix, int rows, int cols){
        int top = 0, bottom = rows - 1;
        int left = 0, right = cols - 1;
        int i;
        System.out.print("\nClockwise Spiral: ");
        while(top <= bottom && left <= right){
            //1. Move right across the top row
            for(i = left; i <= right;i++){
                System.out.print(matrix[top][i]+ " ");
            }
```

```
                //Shrink top boundary
                top++;

                //2. Move Down along the right column
                for(i = top; i <= bottom; i++){
                    System.out.print(matrix[i][right]+ " ");
                }
                //Shrink right boundary
                right--;
                //3. Move left across the bottom row (if rows remain)
                if(top <= bottom){
                    for(i = right; i >= left; i--){
                        System.out.print(matrix[bottom][i] + " ");
                    }
                    //shrink bottom boundary
                    bottom--;
                }

                //4. move up along the left column (if column remain)
                if(left <= right){
                    for(i = bottom; i >= top; i--){
                        System.out.print(matrix[i][left] + " ");
                    }
                    //Shrink left boundary
                    left++;
                }
            }
        }
    }
```